

6. Four models, same structure

Gender compass summary

Model	head	$A(10)$	$\Delta\phi$
GPT-2 Small	L10H9	14.16	0.3°
Phi-3 Mini	L28H1	12.85	1.3°
Gemma-2-2B	L21H4	2.19	1.3°
Llama-3.2-3B	L26H14	3.01	2.4°

- Amplitudes differ by $\sim 5\times$ across families (GPT-2 $\rightarrow$ Gemma).
- Phase is stable everywhere — the dial is always sharp.
- Same two-pillar recipe discovers the plane independently of depth, width, vocab size.